Language Translation Tool

A desktop and terminal app that translates text between 22 languages through a translation API, with one-click copy and text-to-speech built in.

Prepared by · Sunny Gouria

Domain · Artificial Intelligence

Status · completed

CLAIM

Type in one language, read it back in another, in a single round trip.

The source language can be auto-detected; twenty-two languages are offered, including all major Indian languages.

Four features make it usable day to day

01 **A picker for source and target**
Auto-detect the language, or choose from twenty-two.

02 **Live translation through an API**
The public Google Translate endpoint, no API key required.

03 **Copy the result in one click**
The translated text goes straight to the clipboard.

04 **Hear it read aloud**
Optional text-to-speech via pyttsx3 or the system voice.

WHAT IT MEANS

One interface, two ways to run

The same code opens a Tkinter window or runs entirely in the terminal, so it fits wherever you work.

Built with · Python · Tkinter

Source: project README

How a request is translated

01

Input

Text typed in the window or the terminal.

02

Select

Source and target languages chosen.

03

Translate

The API call — the step that owns the latency.

04

Display

Translation shown, ready to copy or speak.

The engine tries the Google endpoint first and falls back automatically, so a rate-limited request still returns a translation.

Source: translator.py

Four parts, one per module

Interface

A Tkinter window with a terminal mode behind the same entry point.

Translator core

All API logic isolated in one file, so the engine can be swapped.

Speech

Text-to-speech with a graceful fallback if the library is missing.

Packaging

requirements.txt and a README so it runs from the VS Code terminal.

Source: project structure

What each capability needs

CAPABILITY	HOW IT WORKS	STATUS	ENGINE
Translate	Public Google endpoint, no key required	HIGH	Google
Auto-detect	Source language inferred by the API	HIGH	Google
Fallback	deep-translator to MyMemory on failure	MEDIUM	MyMemory
Copy to clipboard	Native Tkinter clipboard, no dependency	HIGH	Local
Text-to-speech	pyttsx3 or the operating-system voice	LOW	Local

Copy, auto-detect and translation need nothing beyond Python and the requests the app makes; speech is optional.

Source: requirements.txt

Run it from the VS Code terminal

python main.py opens the window; python main.py --cli runs in the terminal.
Full setup steps are in the README.

TASK 01 · COMPLETE

Built by · Sunny Gouria

Stack · Python · Tkinter

Repo · CodeAlpha_Translate